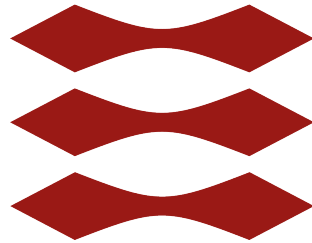


DTU



TECHNICAL UNIVERSITY OF DENMARK

25314 - COMPUTATIONAL MARINE ECOLOGICAL MODELLING

Course Project: Developing NPZDO Model for Sediment Redox Reactions

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1 Introduction

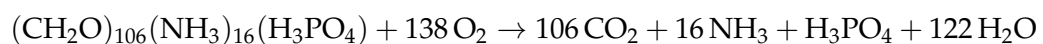
The breakdown of organic matter in marine sediments is governed by a sequence of redox (reduction–oxidation) reactions. As oxygen is rapidly depleted below the sediment-water interface, microorganisms transition to alternative electron acceptors, creating distinct biogeochemical zones that regulate the cycling of nitrogen, phosphorus, and carbon.

The central objective of this study is to determine: **What is the proportion of aerobic decomposition and denitrification in organic matter degradation within the sediments?** By quantifying these pathways, we can better understand the efficiency of nutrient recycling and the extent of anaerobic processing in benthic environments.

Adapting the NPZDO Framework To investigate the proportional contribution of these reactions, we are adapting the NPZDO model originally developed in the first part of this course. While the initial model focused on pelagic dynamics, this version introduces sedimentation to resolve organic matter decomposition over depth. A key modification to our original framework is replacing the general “Nutrient” variable with two distinct state variables: Nitrate (NO_3^-) and Ammonium (NH_4^+).

Chemical Framework The model now accounts for the specific stoichiometry of three primary microbial processes:

- Aerobic Mineralization: The degradation of organic matter using dissolved oxygen



- Denitrification: The degradation of organic matter using nitrate as an electron acceptor under anaerobic conditions



- Nitrification: The aerobic oxidation of ammonium back into nitrate



2 Methodology

To address our research question, we transition the NPZDO framework from a pelagic-only focus to a vertically resolved sediment model (Figure 1). Rather than detailing the standard ecological interactions (grazing, mortality, and uptake) which remain consistent with the original model, this section focuses on the expanded nutrient speciation and the stoichiometric representation of benthic redox reactions.

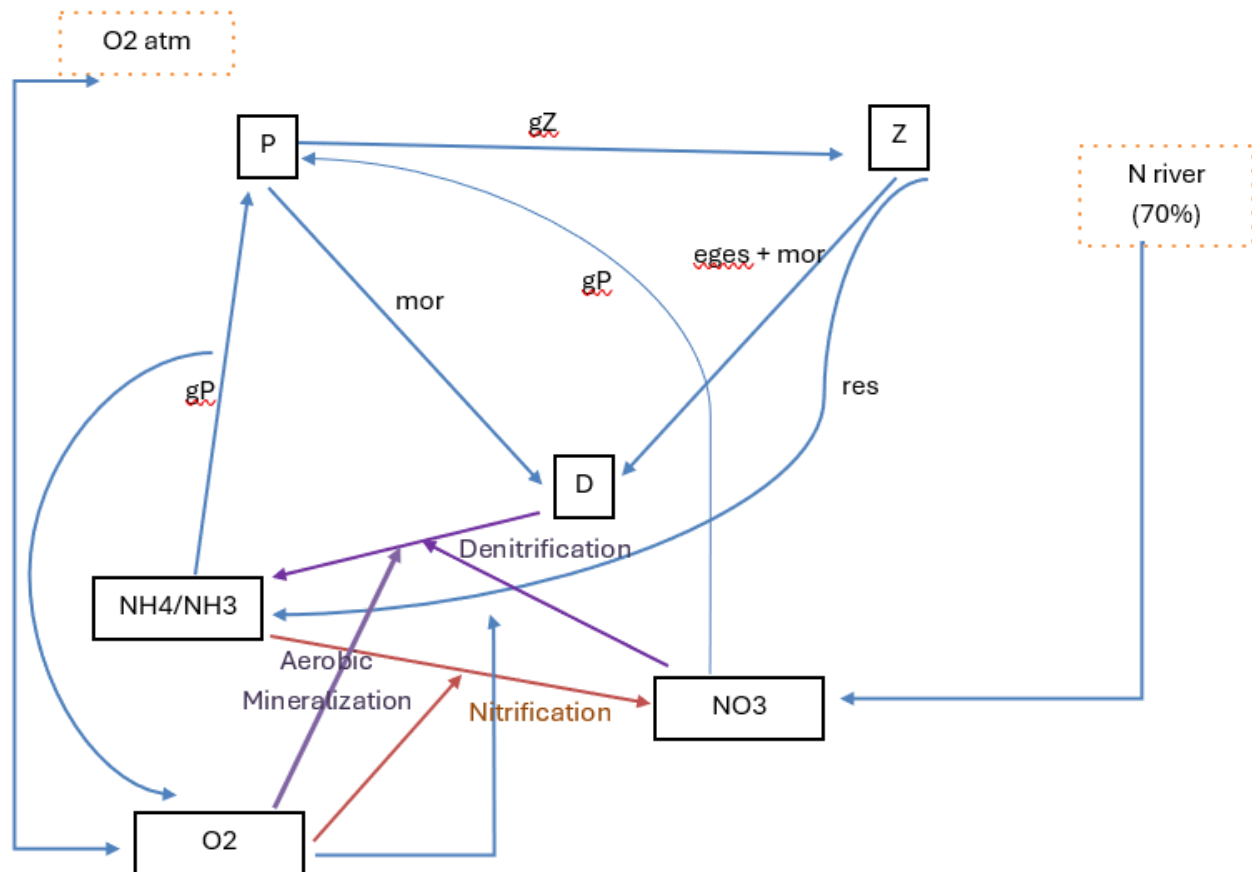


Figure 1: NPZDO Sediment model illustration

2.1 Assumptions

To simplify the complexity of benthic biogeochemical signaling and focus on the primary nitrogen and oxygen cycles, the following assumptions were made in the adaptation of the model:

- **Ammonium Speciation:** For the purposes of this model, NH_3 (ammonia) and NH_4^+ (ammonium) are treated as a single state variable. We do not account for the pH-dependent equilibrium between these two species, as the total reduced inorganic nitrogen is the primary variable of interest for our stoichiometric calculations.

- **Phytoplankton Nutrient Preference:** We assume that phytoplankton use ammonia NH_4^+ and nitrate NO_3^- at the same time when it comes to the nitrogen uptake.
- **Riverine Nitrogen Input:** All nitrogen entering the system via riverine source terms is assumed to be in the form of nitrate (NO_3^-). This is justified by global river export studies showing that nitrate is the dominant inorganic nitrogen species in riverine fluxes, typically accounting for approximately one-third to one-half of total riverine nitrogen export [1].
- **Atmospheric Deposition:** Because the majority of nitrogen input comes from riverine sources, we do not model atmospheric deposition.
- **Redox Sequence Scope:** While we recognize that benthic redox successions include reactions involving manganese, iron, and sulfate, this model is restricted to O_2 and NO_3^- as electron acceptors. We assume these are the dominant drivers for organic matter degradation in the upper sediment layers relevant to our research question.

2.2 Governing Equations

The governing equations for the adapted model are:

$$\frac{\partial \text{NH}_4}{\partial t} = -\frac{\text{lim}_{\text{NH}_4}}{\text{lim}_{\text{NH}_4} + \text{lim}_{\text{NO}_3}} \times g_P \times P + \epsilon_{\text{res}} \times g_Z \times Z - r_{\text{nit}} \times \text{NH}_4 \times \frac{\text{O}_2}{\text{O}_2 + k_{\text{O}_2}} + (\text{OM}_{\text{decomp}, \text{O}_2}(\text{O}_2) + \text{OM}_{\text{decomp}, \text{NO}_3}(\text{O}_2)) \times D + \frac{\partial}{\partial z} \left(\kappa \frac{\partial \text{NH}_4}{\partial z} \right) \quad (1)$$

$$\frac{\partial \text{NO}_3}{\partial t} = -\frac{\text{lim}_{\text{NO}_3}}{\text{lim}_{\text{NH}_4} + \text{lim}_{\text{NO}_3}} \times g_P \times P + r_{\text{nit}} \times \text{NH}_4 \times \frac{\text{O}_2}{\text{O}_2 + k_{\text{O}_2}} - \text{OM}_{\text{decomp}, \text{NO}_3}(\text{O}_2) \times D \times \frac{84.8}{16} + \frac{\partial}{\partial z} \left(\kappa \frac{\partial \text{NO}_3}{\partial z} \right) \quad (2)$$

$$\frac{\partial P}{\partial t} = g_P \times P - g_Z \times Z - m_P \times P + \frac{\partial}{\partial z} \left(\kappa \frac{\partial P}{\partial z} \right) \quad (3)$$

$$\frac{\partial Z}{\partial t} = (1 - \epsilon_{\text{res}} - \epsilon_{\text{eges}}) \times g_Z \times Z - m_Z \times Z^2 + \frac{\partial}{\partial z} \left(\kappa \frac{\partial Z}{\partial z} \right) \quad (4)$$

$$\frac{\partial D}{\partial t} = m_P \times P + m_Z \times Z + \epsilon_{\text{eges}} \times g_Z \times Z - (\text{OM}_{\text{decomp}, \text{O}_2}(\text{O}_2) + \text{OM}_{\text{decomp}, \text{NO}_3}(\text{O}_2)) \times D + \frac{\partial}{\partial z} \left(\kappa \frac{\partial D}{\partial z} \right) - w_D \times \frac{\partial D}{\partial z} \quad (5)$$

$$\frac{\partial \text{O}_2}{\partial t} = \frac{138}{16} \times g_P \times P - \frac{106}{16} \times \epsilon_{\text{res}} \times g_Z \times Z - \frac{106}{16} \times \text{OM}_{\text{decomp}, \text{O}_2}(\text{O}_2) \times D - 2 \times r_{\text{nit}} \times \text{NH}_4 \times \frac{\text{O}_2}{\text{O}_2 + k_{\text{O}_2}} + \frac{\partial}{\partial z} \left(\kappa \frac{\partial \text{O}_2}{\partial z} \right) \quad (6)$$

2.2.1 Organic Matter Degradation

To switch from aerobic decomposition, we defined a function called OM_{decomp, O_2} :

$$OM_{\text{decomp}, O_2}(O_2) = \begin{cases} \tau(O_2)r_{O_2}\frac{O_2}{O_2+k_{O_2}}, & O_2 \geq O_{2,\text{lim}} \\ 0, & \text{else} \end{cases} \quad (7)$$

Denitrification:

$$OM_{\text{decomp}, NO_3}(O_2) = \begin{cases} 0, & O_2 \geq O_{2,\text{lim}} \\ r_{\text{denit}}\frac{NO_3}{NO_3+k_{NO_3}}, & \text{else} \end{cases} \quad (8)$$

2.2.2 Phytoplankton Growth (g_P)

Phytoplankton growth is limited by both light intensity (I) and nutrient availability. In this adapted model, the original unified "Nutrient" state variable has been replaced by NH_4^+ and NO_3^- . Therefore, the nutrient limitation for phytoplankton growth is defined as below [2]:

$$g_P = g_{P,\text{max}} \cdot \min \left(\frac{I}{I + k_L}, \frac{NO_3 e^{-2NH_4}}{NO_3 + k_N} + \frac{NH_4}{NH_4 + k_N} \right) \quad (9)$$

2.2.3 Nutrient limitation ($\lim_{NH_4}$ and $\lim_{NO_3}$)

To balance nutrient uptake from ammonium to nitrate, we define the following functions:

$$\lim_{NH_4} = \frac{NH_4}{NH_4 + k_N}, \quad \lim_{NO_3} = \frac{NO_3 e^{-2NH_4}}{NO_3 + k_N} \quad (10)$$

2.2.4 Grazing and Remineralization

Zooplankton consumption g_Z , aerobic mineralization rate r_{O_2} and denitrification rate r_{denit} are all oxygen-dependent via a function $\tau(O_2)$:

$$g_Z(O_2) = g_{Z,\text{max}} \frac{P}{P + k_Z} \tau(O_2), \quad \tau(O_2) = \frac{1}{2} \left(1 + \tanh \left(\frac{O_2 - 100}{10} \right) \right) \quad (11)$$

2.2.5 Seasonality and Physical Forcing

While the biological and chemical components of the model have been expanded, the physical and environmental forcing remains consistent with the original framework. Specifically, the seasonal diffusivity ($\kappa(z, t)$) and seasonal light intensity ($I(z, t)$) remain unchanged.

2.3 Modifications and New Dynamics

The primary evolution of the model lies in the treatment of the detritus (D) pool and inorganic nitrogen. We have implemented several critical changes:

- **Monod Kinetics for Redox Pathways:** The degradation of detritus is now split between aerobic mineralization and denitrification. We utilize Monod-style limitation terms ($\frac{O_2}{O_2+k_{O_2}}$ and $\frac{NO_3}{NO_3+k_{NO_3}}$) to determine which pathway is active based on the local availability of electron acceptors.
- **Nitrogen Speciation:** The original singular nutrient variable is bifurcated into NH_4 and NO_3 . This allows us to explicitly model *nitrification* as well as the specific nitrate demand during anaerobic respiration (denitrification).
- **Stoichiometric Scaling:** In the NO_3 equation, the term $\frac{84.8}{16}$ is applied to the denitrification sink to maintain stoichiometric consistency with the Redfield ratio, accounting for the specific moles of nitrate required per mole of organic nitrogen remineralized. Similarly, in the O_2 equation, several coefficients ($\frac{138}{16}$, $\frac{106}{16}$, 2) are applied to account for the oxygen demand of oxidizing ammonium.
- **Benthic Transport:** All variables continue to be subject to vertical diffusion (κ), while the detritus D include a sinking velocity (w) to represent sedimentation through the water column and into the sediment layers.

2.4 Parameters

Parameters inherited from the original NPZDO model were maintained at their initial values. We introduced several new parameters, including reaction rates, half-saturation constants, the sediment diffusivity coefficient, and nitrogen loading rates from riverine sources.

Table 1: Model Parameters for NPZDO-Sediments Dynamics

Parameter Description	Symbol	Value	Unit
<i>Water Column</i>			
Water column depth	H_W	15.0	m
Sediment depth	H_S	1.0	m
Thickness of grid cells in water column	ΔZ_W	0.5	m
Thickness of grid cells in sediment	ΔZ_S	0.1	m
Number of cells in water column	n_W	30	–
Number of cells in sediment	n_S	10	–
<i>Diffusivity</i>			
Upper-ocean diffusivity	κ_0	4.32	m ² /day
Deep diffusivity	$\kappa_{\min}$	0.432	m ² /day
Sediments diffusivity coefficient	κ_{sed}	0.009	m ² /day
Thermocline depth (mixed layer depth)	$z_0, z_{\text{MixSummer}}$	7.5	m
Winter mixed-layer depth	$z_{\text{MixWinter}}$	13.5	m
Transition thickness	ζ	1.0	m
<i>Light & Seasonality</i>			
Default surface light intensity (in summer)	$I_0, I_{0,\text{summer}}$	70.0	ph/m ² /s
Light atten. (water)	k_W	0.2	m ⁻¹
Light atten. (plankton)	k_P	0.05	m ² /mmol N
<i>Phytoplankton</i>			
Max growth rate	$g_{P,\max}$	1.5	day ⁻¹
Light half-saturation	k_L	4.0	μmol ph/m ² /s
Mortality rate	m_P	0.1	day ⁻¹
Nutrient half-saturation	k_N	0.5	mmol/m ³
<i>Zooplankton</i>			
Max grazing rate	$g_{Z,\max}$	0.7	day ⁻¹
Half-sat. grazing	k_Z	2.0	mmol/m ³
Respiration fraction	ϵ_{res}	0.3	-
Egestion fraction	ϵ_{eges}	0.3	-
Mortality rate	m_Z	0.1	day ⁻¹ / mmol / m ³
<i>Detritus</i>			
Sinking speed	w_D	5.0	m/day
Oxygen threshold for aerobic mineralization	$O_{2,\text{lim}}$	5.0	mmol O ₂ /m ³
Aerobic mineralization rate	r_{O_2}	0.1	day ⁻¹
Denitrification rate	r_{denit}	0.2	day ⁻¹
Oxygen half-saturation	k_{O_2}	10.0	mmol/m ³
Nitrate half-saturation	k_{NO_3}	1.0	mmol/m ³
<i>Oxygen</i>			
Atmospheric O ₂ saturation	$O_{2,\text{atm}}$	260.0	mmol O ₂ /m ³
Gas transfer velocity at sea surface	g_{tv}	8.64	m/day
<i>Nitrification</i>			
Nitrification rate	r_{nit}	0.1	day ⁻¹
Ammonium/nitrate threshold for photosynthesis	$NH_{4,\text{lim}}$	0.1	mmol NH ₄ /m ³
River NO ₃ run-off	N_{river}	0.5	mmol/m ² /day

2.5 Numerical scheme

2.5.1 Spatial Discretization

The model domain consists of a water column of depth $H_W = 15$ m and a sediment layer of depth $H_S = 1$ m, discretized into $n_W = 30$ and $n_S = 10$ cells, respectively (Figure 2).

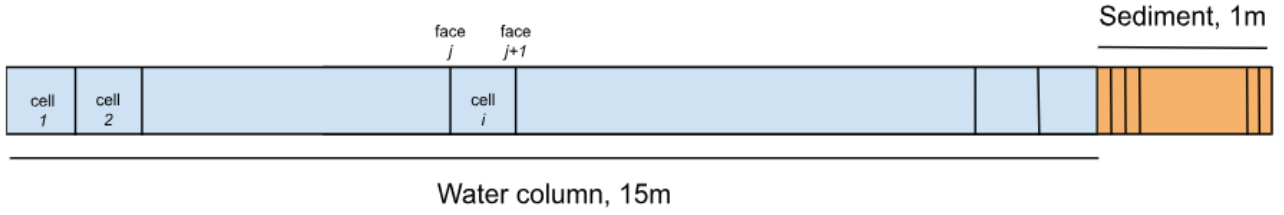


Figure 2: Spatial discretization sketch

The thickness of each layer is defined as:

$$\Delta z_i = \begin{cases} \Delta z_W = 0.5 \text{ m}, & 0 \leq i < n_W \\ \Delta z_S = 0.1 \text{ m}, & n_W \leq i < n_W + n_S \end{cases} \quad (12)$$

The concentration C_i is defined at the center of each cell:

$$z_i = \begin{cases} (i + 0.5)\Delta z_W, & 0 \leq i < n_W \\ H_W + (i - n_W + 0.5)\Delta z_S, & n_W \leq i < n_W + n_S \end{cases} \quad (13)$$

Fluxes J_i are defined at the interfaces between cells, where the index i ranges from 0 to $n_W + n_S$. The discrete conservation equation for each cell is given by:

$$\frac{dC_i}{dt} = -\frac{J_{i+1} - J_i}{\Delta z_i} + \text{Sources/Sinks}_i. \quad (14)$$

At the sediment-water interface ($z = H_W$), the flux continuity is maintained to ensure the exchange of dissolved species between the two domains.

2.5.2 Advection and diffusion

The total flux J is the sum of advective and diffusive components. We use the same numerical scheme as in the original NPZDO model.

The advective flux J_a is discretized using an upwind scheme:

$$J_{a,i} = wC_{i-1}, \quad (15)$$

where w represents the sinking speed.

The diffusive flux J_d is discretized using a central difference scheme:

$$J_{d,i} = -\kappa_i \frac{C_i - C_{i-1}}{\Delta z}, \quad (16)$$

where κ_i is the diffusion coefficient.

2.5.3 Boundary Conditions

With the transition from a pelagic-only domain to a coupled water-sediment model, the boundary conditions have been configured to ensure mass conservation within the extended system. Living phytoplankton and zooplankton are only present in the water column, but the other state variables have both a pelagic and a benthic value, hence their decomposition into two arrays. For instance, when looking at ammonium, $NH_{4W_{n_W}}$ is the last cell in water column and the nearest one to seabed ; while NH_{4S_0} is the first cell in sediment and the nearest one to seabed.

Variable	Sea-Air Flux	Seabed Flux	Seafloor Flux
Phytoplankton (P)	0	0	–
Zooplankton (Z)	0	0	–
Water Column Ammonium (NH_{4W})	0	$-\kappa_{\min} \frac{NH_{4S_0} - NH_{4W_{n_W}}}{\frac{\Delta z_W}{2} + \frac{\Delta z_S}{2}}$	–
Sediment Ammonium (NH_{4S})	–	$-\kappa_{\min} \frac{NH_{4S_0} - NH_{4W_{n_W}}}{\frac{\Delta z_W}{2} + \frac{\Delta z_S}{2}}$	0
Water Column Nitrate (NO_{3W})	N_{river}	$-\kappa_{\min} \frac{NO_{3S_0} - NO_{3W_{n_W}}}{\frac{\Delta z_W}{2} + \frac{\Delta z_S}{2}}$	–
Sediment Nitrate (NO_{3S})	–	$-\kappa_{\min} \frac{NO_{3S_0} - NO_{3W_{n_W}}}{\frac{\Delta z_W}{2} + \frac{\Delta z_S}{2}}$	0
Water Column Detritus (D_W)	0	$-\kappa_{\min} \frac{D_{S_0} - D_{W_{n_W}}}{\frac{\Delta z_W}{2} + \frac{\Delta z_S}{2}}$	–
Sediment Detritus (D_S)	–	$-\kappa_{\min} \frac{D_{S_0} - D_{W_{n_W}}}{\frac{\Delta z_W}{2} + \frac{\Delta z_S}{2}}$	0
Water Column Oxygen (O_{2W})	$-\text{gtv}(O_{2W_0} - O_{2,\text{atm}})$	$-\kappa_{\min} \frac{O_{2S_0} - O_{2W_{n_W}}}{\frac{\Delta z_W}{2} + \frac{\Delta z_S}{2}}$	–
Sediment Oxygen (O_{2S})	–	$-\kappa_{\min} \frac{O_{2S_0} - O_{2W_{n_W}}}{\frac{\Delta z_W}{2} + \frac{\Delta z_S}{2}}$	0

Table 2: Boundary flux expressions

3 Results

3.1 Mesh Dependency

It is necessary to check if the results are independent of the mesh resolution. For this aim, 50-day simulations were conducted with different values of cell thickness both in water column and sediment. Figure 3 demonstrates how the phytoplankton concentration at the middle of the water column, depth = 7 m, changes with size of grid cells in water column. Based on this figure, the values of 0.5 and 0.025 m have been used for the results in Section 3.3.

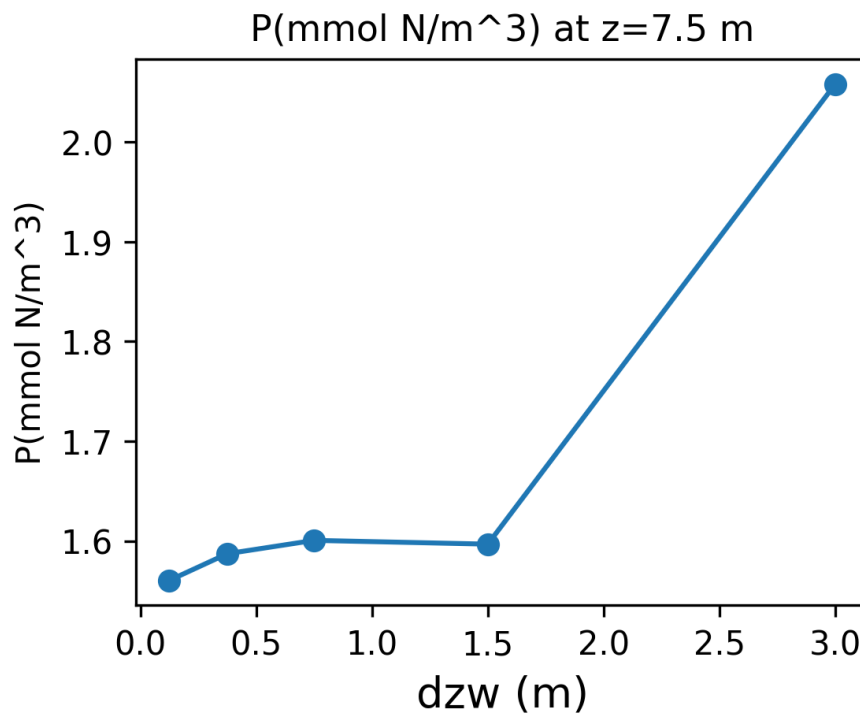


Figure 3: Phytoplankton concentration at $z = 7.5$ m does not change considerably for water column cell thickness below 1.5 m

3.2 Convergence

To realize when the system becomes converged with respect to initial and boundary conditions and only reflects seasonality, a 5-year simulation was run. According to the results in Figure 4, the system reaches convergence after 3 years. Therefore, in Section 3.3, the results of a 3-year simulation are discussed.

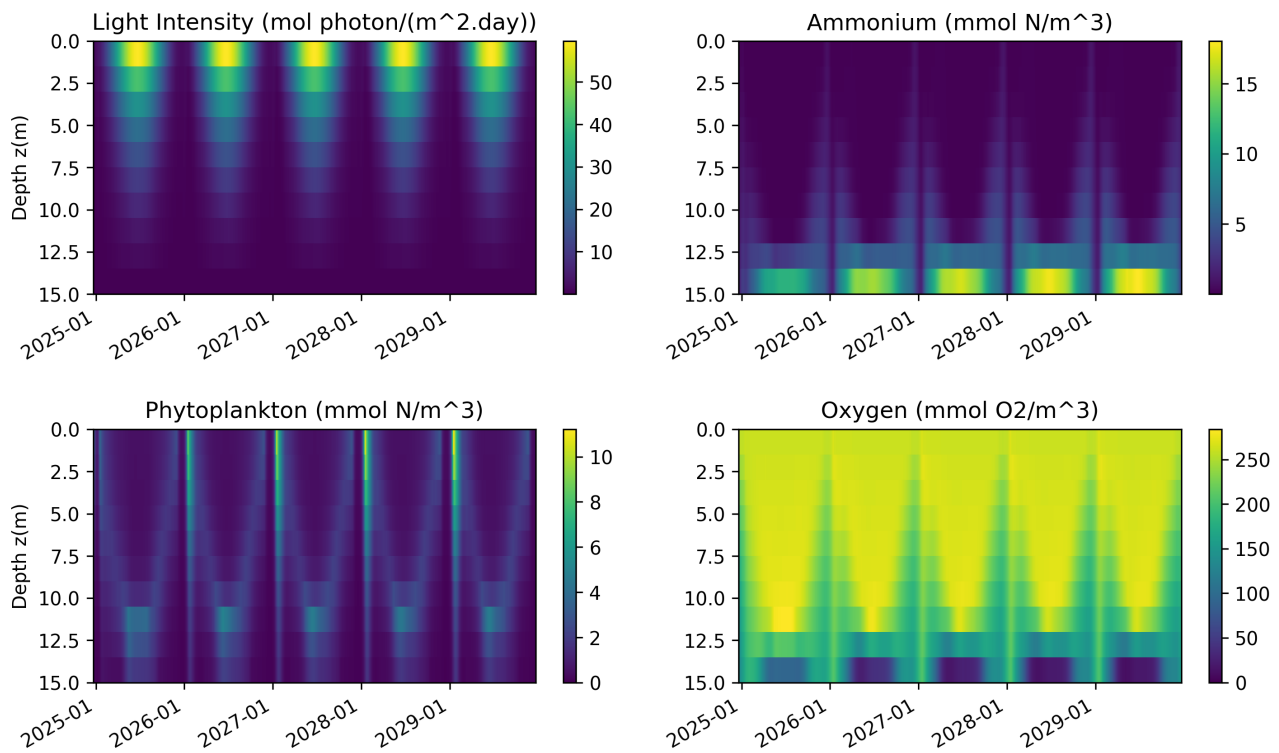


Figure 4: After 3 years, the system converges with respect to initial and boundary conditions, reflecting only seasonality.

3.3 Summer and Winter Conditions

Figure 5 demonstrates how light irradiance changes at the surface and inside the water throughout the year, i.e. summer and winter conditions.

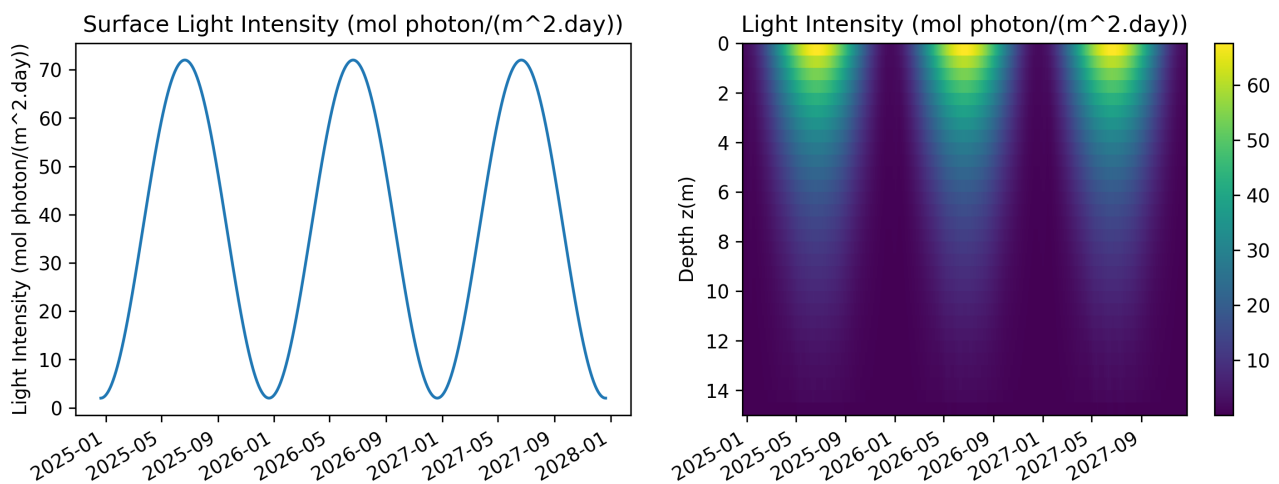


Figure 5: Seasonal changes in light irradiance at the surface and inside the water.

Figure 6 illustrates the vertical profiles of different parameters and state variables throughout the water column, at a timepoint in winter when surface light is the minimum. In this condition, primary production is always limited by light, but not by nutrient. That results in the maximum concentration of phytoplankton at the surface. Owing to strong winds and waves at this time of the year, the diffusivity coefficient is high in most of the water column and a well-mixed pattern is observed. Moreover, some detritus accumulates on the seabed, increasing the rate of aerobic respiration. On the other hand, since oxygen is available, the rate of denitrification is zero throughout the water column. This oxygen also turns ammonium into nitrate, leading to higher concentrations for the latter.

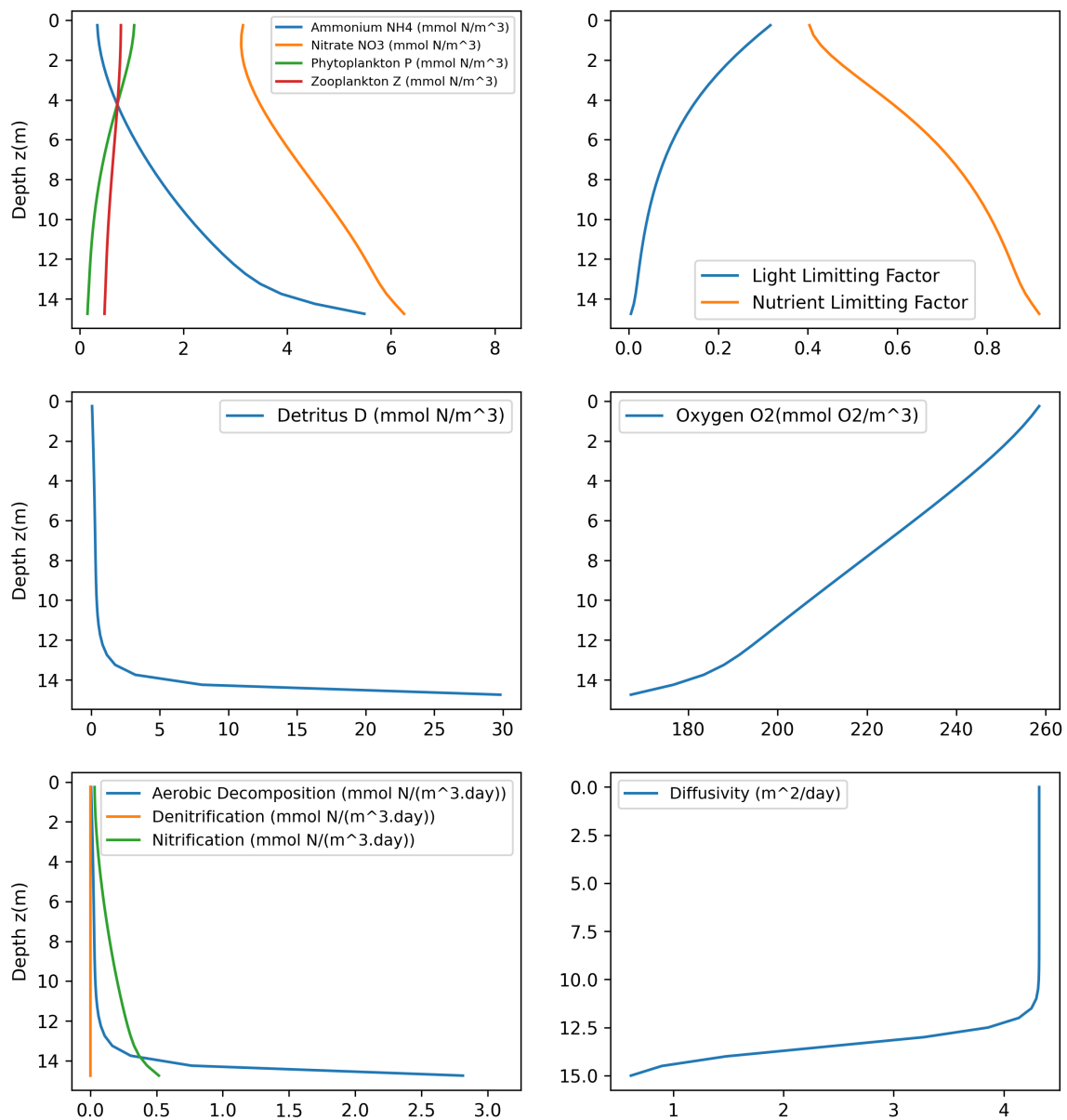


Figure 6: Vertical profiles of ammonia, nitrate, phytoplankton, zooplankton, light and nutrient limitations, detritus, oxygen, rate of aerobic or anaerobic detritus decomposition, rate of nitrification, and diffusivity in winter.

In summer, as shown in Figure 7, although light is quite available, nutrients are limited. Consequently, phytoplankton peak deeper in water at about 12 meters. The greater photosynthesis at this time leads to higher detritus sinking on seafloor. The uniform vertical profile of oxygen in the upper water column represents the well-mixed layer, followed by a peak due to oxygen produced by phytoplankton.

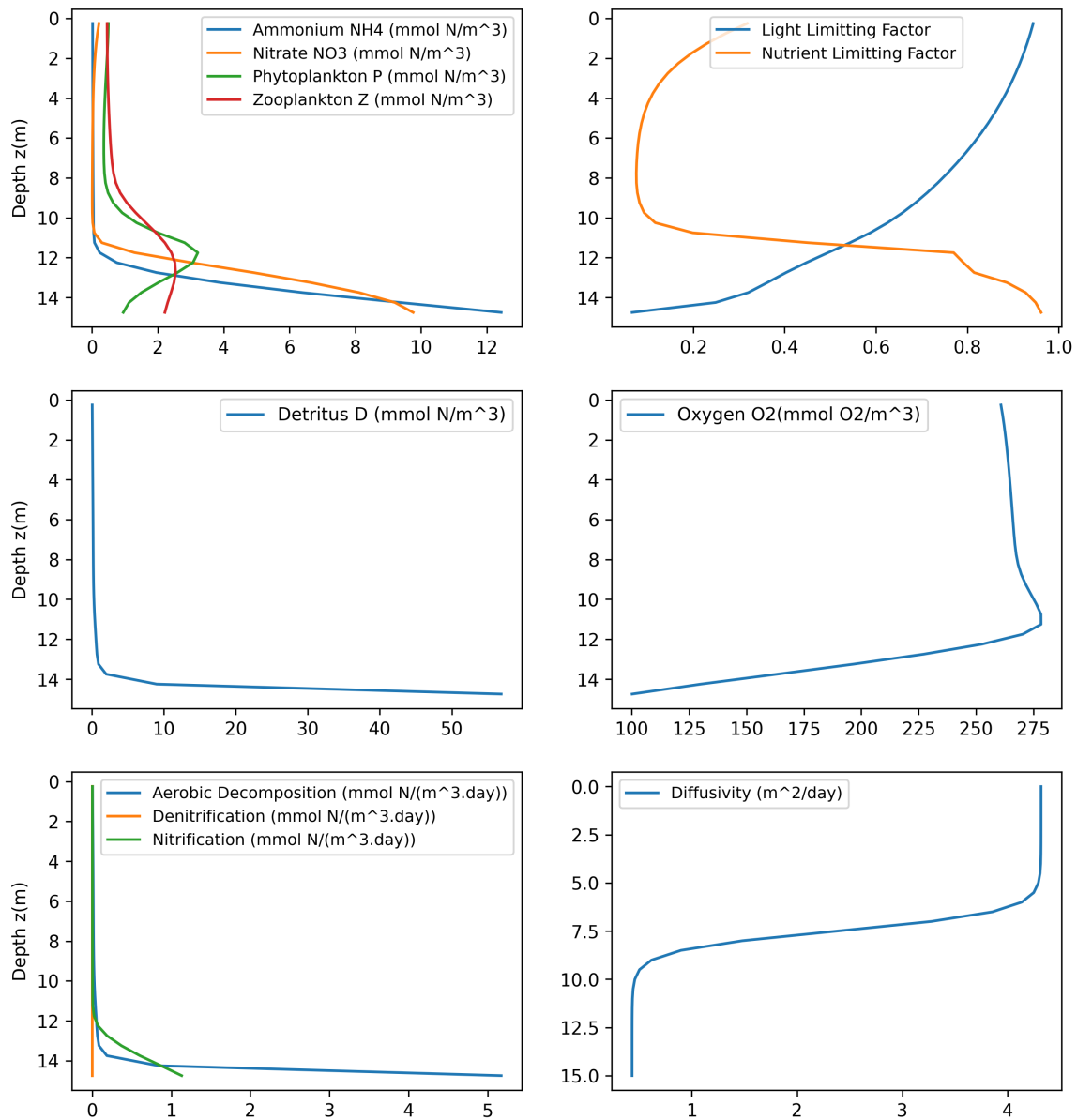


Figure 7: Vertical profiles of ammonia, nitrate, phytoplankton, zooplankton, light and nutrient limitations, detritus, oxygen, rate of aerobic or anaerobic detritus decomposition, rate of nitrification, and diffusivity in summer.

Similarly, it is shown in Figure 8 that phytoplankton stay near the surface at winter and move to the deep in summer, and zooplankton chase them. During spring and summer, owing to higher primary production, more detritus accumulates on the seafloor compared to that in winter. This

is the reason behind lower oxygen levels at the bottom during summer, as oxygen is required for detritus decomposition. As for nutrients, while in summer, larger magnitudes of ammonium and nitrate are recycled from the organic matter deposited on the seabed, these nutrients can hardly diffuse to shallower depths due to high uptake by phytoplankton.

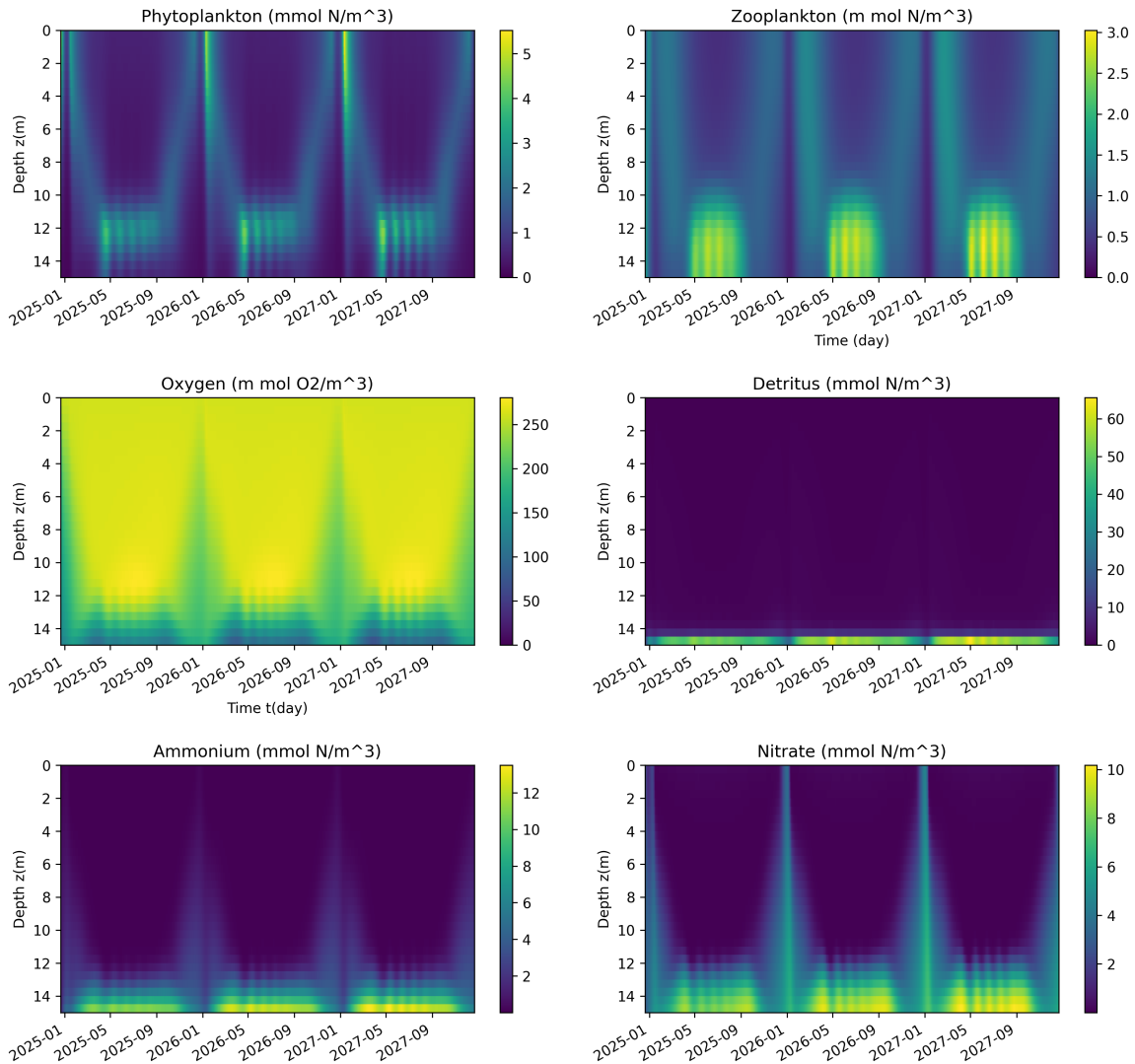


Figure 8: Concentration of phytoplankton, zooplankton, oxygen, detritus, ammonium, and nitrate throughout the water column over time.

The detritus which builds up on the seabed diffuses into the sediment. There, it is mineralized initially by oxygen and when oxygen gets depleted, by nitrate. As photosynthesis is stronger in summer, detritus concentrations are higher, as shown in Figure 9. This demands more oxygen for detritus decomposition and that is why the maximum oxygen value is lower in summer. Furthermore, it is observed that in summer, the oxygen gets exhausted at a shallower depth (faster) at about 0.3 m deep in sediments compared to the 0.5 m equivalent in winter. This higher rate of organic matter degradation is also visible in denitrification rate maximum at above $3 \text{ mmolN.m}^{-3}.\text{day}^{-1}$ in summer compared to the $1.5 \text{ mmolN.m}^{-3}.\text{day}^{-1}$ in winter. In both cases, detritus concentration decreases with depth

until the oxidants, oxygen and nitrate, are drained, followed by a slight increase before leveling out at a fixed value.

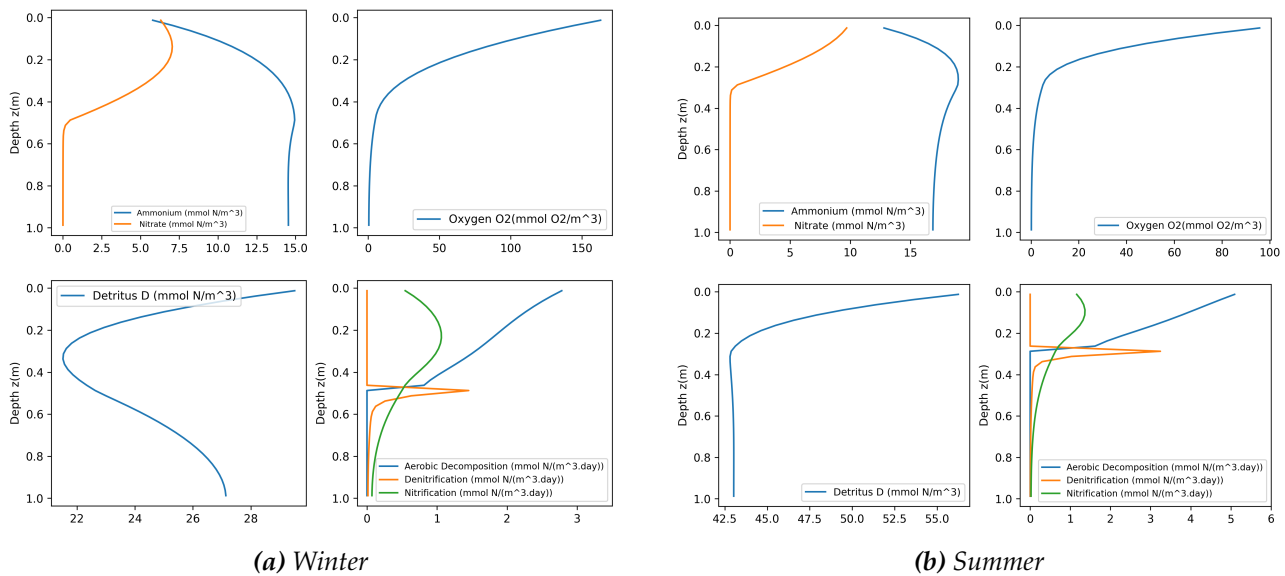


Figure 9: Vertical profiles of ammonium, nitrate, oxygen, detritus and rates of aerobic respiration, denitrification and nitrification in the sediments during winter and summer 2027.

The same interpretations as the previous paragraph can be made by looking at the concentration field of oxygen and nitrate, and the rates of aerobic respiration and denitrification in Figure 10.

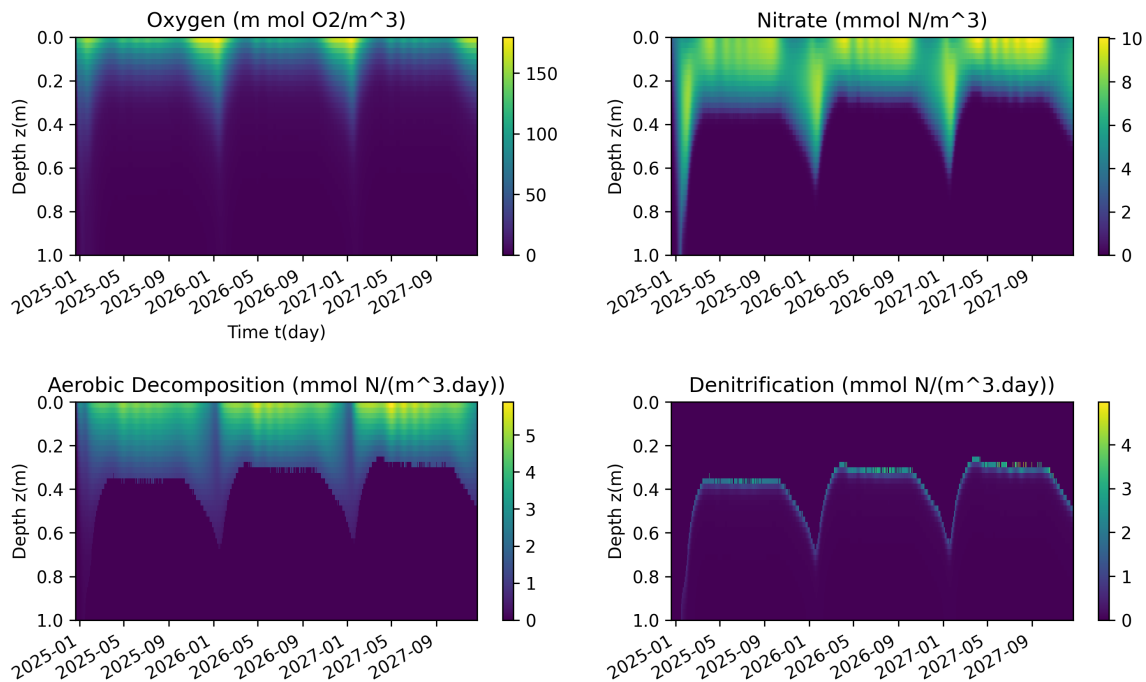


Figure 10: Concentration of oxygen and nitrate and rates of aerobic respiration and denitrification in the sediments over time.

Figure 11 shows the proportion of aerobic respiration in total detritus degradation. It is clearly seen that oxygen is the main oxidant. However, in summer when denitrification is higher in the sediments than winter, there is a minimal decrease in aerobic contribution. In denitrification reaction, nitrate turns into dinitrogen N_2 bubbles, moves up into the water column, and goes into the atmosphere. Figure 12 demonstrates the rate of N_2 generation which is a sink for the nitrogen cycle. This rate is lightly higher in summer than in winter.

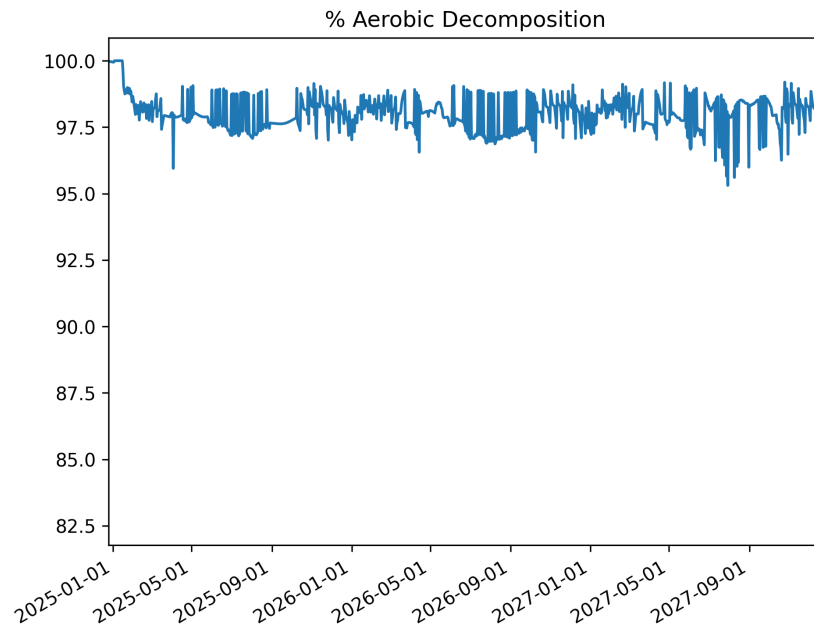


Figure 11: Percentage of aerobic organic matter degradation in total decomposition.

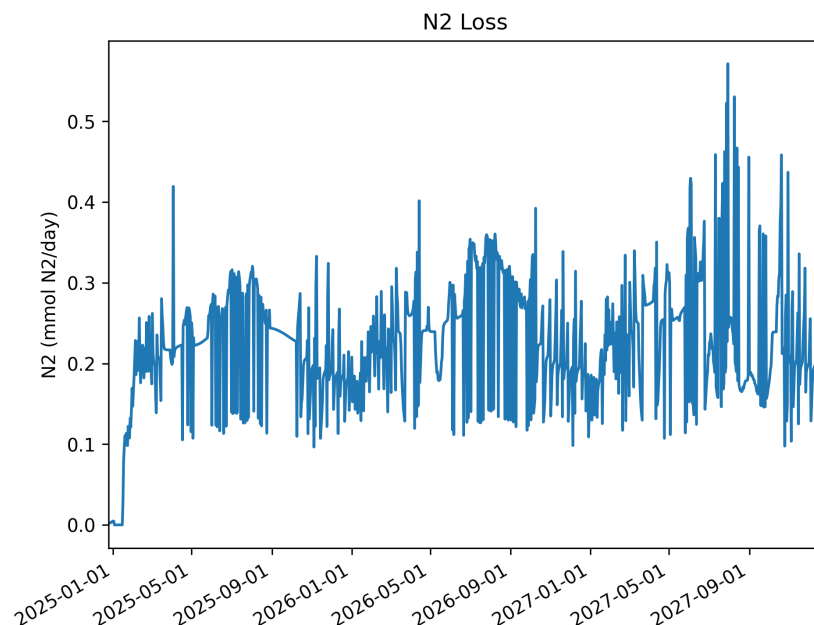


Figure 12: Rate of N_2 generation in the denitrification process, which is a sink in the marine nitrogen cycle.

4 Conclusion

Although oxygen is the dominant oxidant for detritus decomposition in both water column and sediments, denitrification acts a supporting role where oxygen is used up in the sediments.

5 Link to our code on GitHub

You can find our code on GitHub at https://github.com/disco00o/25314-NPZD0_Sediments.

References

- [1] Elizabeth Boyer and Robert Howarth. "Nitrogen Fluxes from Rivers to the Coastal Oceans". In: Dec. 2008, pp. 1565–1587. ISBN: 9780123725226. DOI: 10.1016/B978-0-12-372522-6.00036-0.
- [2] M. J. R. Fasham, H. W. Ducklow, and S. M. McKelvie. "A nitrogen-based model of plankton dynamics in the oceanic mixed layer". In: *Journal of Marine Research* 48.3 (1990), pp. 591–639. DOI: 10.1357/002224090784984678.